Reg. No. | | | | | | | | | | | [18AS732]

# SEVENTH SEMESTER B.Tech EXAMINATION, DECEMBER–2021
## AIRCRAFT SYSTEMS

**Time: 3 Hours** **Maximum Marks: 70**

**Instructions:**
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

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| **1.** | **a.** | Explain the need for powered mechanical flight control systems. | **05** |
| | **b.** | Explain the functioning of a fly-by-wire system. | **09** |
| **2.** | **a.** | Write short notes on the following: | |
| | | **i.** Fuel booster pumps | **02** |
| | | **ii.** Cavitation | **02** |
| | | **iii.** Smart probes in fuel quantity measurement system | **03** |
| | **b.** | Discuss the various fuel level sensors used in aircraft fuel systems with neat diagrams wherever available. | **07** |
| **3.** | **a.** | Discuss the hydraulic System application in BAE SYSTEMS Hawk 200 aircraft. | **07** |
| | **b.** | Explain the method of AC power generation in aircraft. | **07** |
| | | **OR** | |
| **4.** | **a.** | Explain in brief the simplified Airbus A320 Hydraulic system. | **07** |
| | **b.** | Write short notes on: | |
| | | **i.** Aircraft electrical batteries | **04** |
| | | **ii.** Aircraft circuit breakers | **03** |
| **5.** | **a.** | Write short notes on: | |
| | | **i.** Ice protection / detection systems | **03** |
| | | **ii.** Engine start using pneumatic power | **04** |
| | **b.** | Discuss the effects of icing on aircraft parts while in flight. Explain in brief the various anti-icing/de-icing methods. | **07** |
| | | **OR** | |
| **6.** | **a.** | Discuss in detail the methods of thrust reversing employed in aircraft. | **07** |
| | **b.** | Describe the phenomenon of Hypoxia and elaborate on how it affects flying in an airplane. | **07** |
| **7.** | **a.** | State the the eight most common Non Destructive Testing methods . | **05** |
| | **b.** | Discuss the principle of ultrasonic testing in NDT of aircraft components. At what stage of an aircraft engine overhaul is ultrasonic test carried out?. | **09** |
| | | **OR** | |
| **8.** | **a.** | How does the eddy current NDT on aircraft components work? At what stage of an aircraft engine overhaul can eddy current test be carried out? | **07** |
| | **b.** | Discuss the various methods of magnetisation of a specimen for the magnetic particle testing. | **07** |